

Solve

ID:

Name:

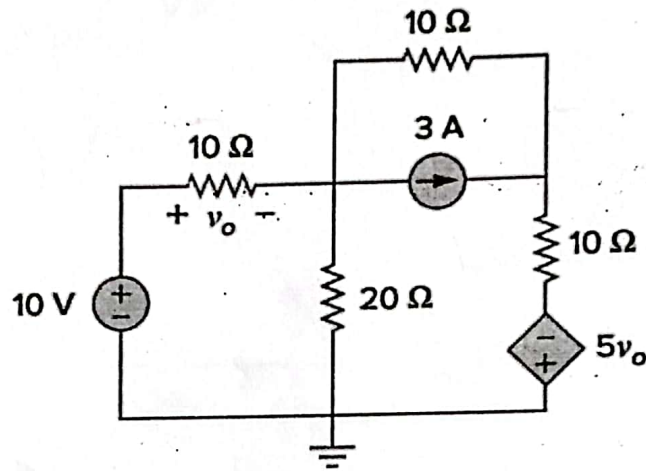
Brac University
Semester: Spring 2025
Course Code: CSE250
Circuits and Electronics
Section: 07

Assessment: Assignment 1
CO3
Deadline: 16.03.25; 11:59 pm
Full Marks: 20

Question 1 of 4

[5 marks]

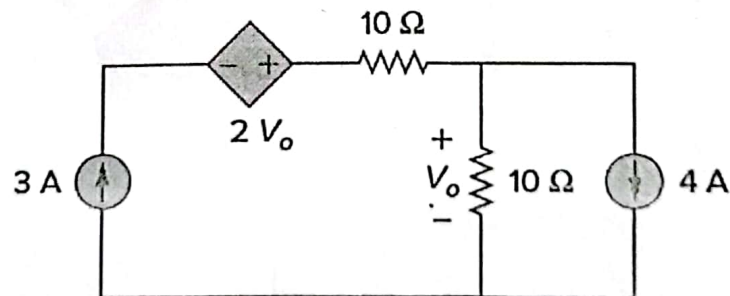
Using nodal analysis, determine v_o in the circuit below.



Question 2 of 4

[5 marks]

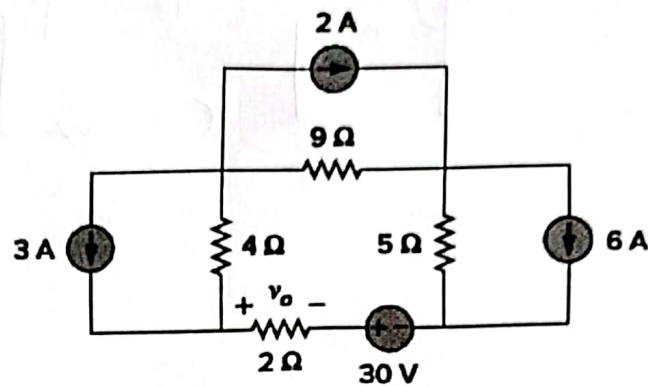
Using nodal analysis to solve for the power of 10 Ω resistor.



Question 3 of 4

[5 marks]

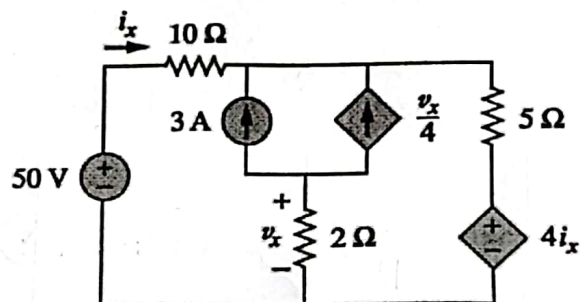
Obtain v_o in the following circuit using mesh analysis.

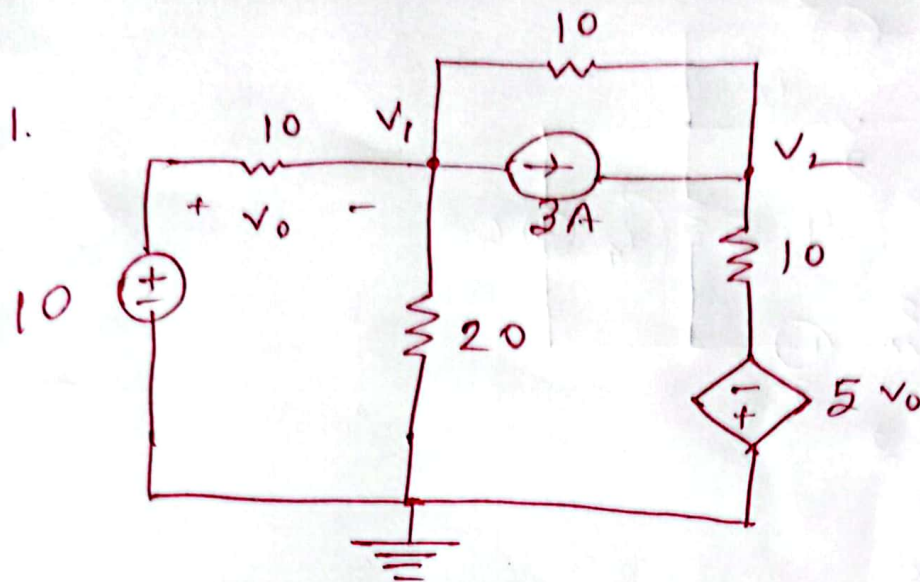


Question 4 of 4

[5 marks]

Find the power of the 3A current source. Use mesh analysis for solving the circuit.





dependent source,

$$v_0 = 10 - v_1$$

at node ①,

$$\frac{v_1 - 10}{10} + \frac{v_1}{20} + 3 + \frac{v_1 - v_2}{10} = 0$$

$$\Rightarrow 2v_1 - 20 + v_1 + 60 + 2v_1 - 2v_2 = 0$$

$$\Rightarrow 5v_1 - 2v_2 = -40 \rightarrow \textcircled{1}$$

at node ②

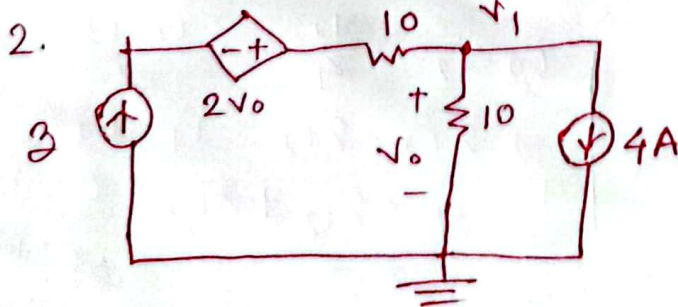
$$-3 + \frac{v_2 + 5v_0}{10} + \frac{v_2 - v_1}{10} = 0$$

$$\Rightarrow -30 + v_2 + 50 - 5v_1 + v_2 - v_1 = 0$$

$$\Rightarrow -6v_1 + 2v_2 = -20 \rightarrow \textcircled{2}$$

$$v_0 = 10 -$$

$$v_1 = 60V, \quad v_2 = 170V$$



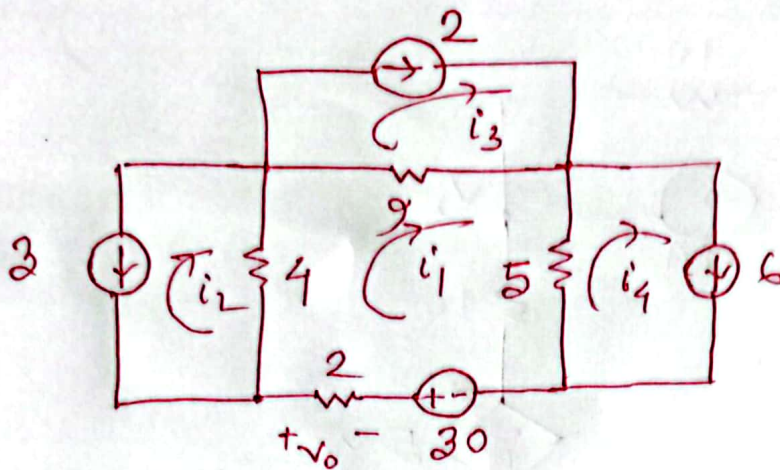
Here, $v_1 = v_0$

$$-3 + \frac{v_1}{10} + 4 = 0$$

$$v_1 = -10V$$

$$P = \frac{v_1^2}{10} = 10W$$

3.



Here, $i_2 = -3$

$i_3 = 2$

$i_4 = 6$

At mesh ①,
 $-30 + 2i_1 + 4(i_1 - i_2) + 9(i_1 - i_3) + 5(i_1 - i_4) = 0$

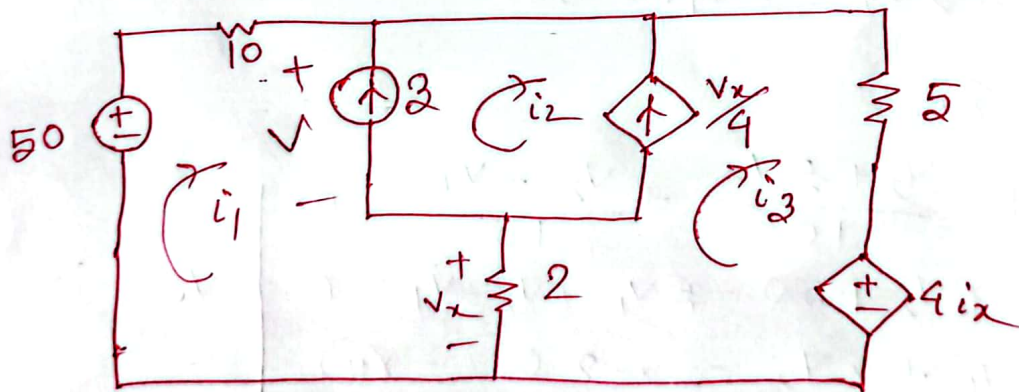
$\Rightarrow -30 + 2i_1 + 4i_1 + 12 + 9i_1 - 18 + 5i_1 - 30 = 0$

$\Rightarrow 20i_1 - 66 = 0$

$i_1 = \frac{66}{20}$

$v_0 = -2i_1 = -2 \times \frac{66}{20} = -6.6 \text{ V}$

4.



$v_x = 2(i_1 - i_3), \quad i_x = i_1$

Now,

$i_2 - i_1 = 3$

$\Rightarrow -i_1 + i_2 = 3 \quad \text{--- (i)}$

$i_3 - i_2 = \frac{v_x}{4} = \frac{i_1 - i_3}{2}$

$\Rightarrow 2i_3 - 2i_2 = i_1 - i_3$

$\Rightarrow -i_1 - 2i_2 + 3i_3 = 0 \quad \text{--- (ii)}$

at, supermesh ①, (ii), (iii)

$-50 + 10i_1 + 5i_3 + 4i_x = 0$

$\Rightarrow 10i_1 + 5i_3 + 4i_1 = 50$

$\Rightarrow 14i_1 + 5i_3 = 50 \quad \text{--- (iii)}$

KVL at left, $-50 + 10i_1 + v + 2(i_1 - i_3) = 0$

$\Rightarrow v = 49.5 + 32.95$

$P = -v \times 3 = -98.85 \text{ W}$